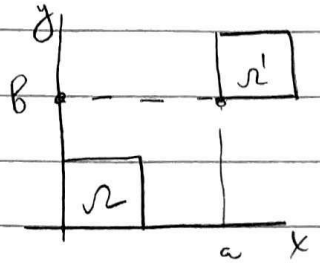
Prob. 1.

Show that if $u(x,y)$ is a harmonic f-n in Ω, then $u(x+a, y+b)$ is harmonic in the dom-n

(1) $$\Omega' = \{ (x+a), (y+b) : (x,y) \in \Omega \}$$

A sample Ω & Ω' are given as:

(2)



This shows that if u is harmonic in a dom-n, & the dom-n is translated, then translating the coordinates gives a harmonic f-n in the new domain.

Sol-n. Let $u(x,y)$ be harmonic in Ω. $\Rightarrow$ By the def-n of harmonic f-ns,

(3) $$u \in C^2(\Omega) \quad \& \quad \Delta u = 0.$$

By the hint given before the pr-s, we must check 2 things:

(i) Change of variable coordinate fr-n is well-defined ($J \neq 0$)

(ii) Laplace eq-n does NOT change.

Then

(i) Let

(4) $$\xi = x + a \quad \& \quad \eta = y + b, \quad \text{then}$$

$$J = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{the transf-n in (4)}$$

is well-defined (works).

$\xi = x + a$
$\eta = y + b$

(ii). Also by the previous HW. we know

$$\begin{cases} u_{xx} = u_{\xi\xi}(\xi_x)^2 + 2u_{\xi\eta}\xi_x\eta_x + u_{\eta\eta}(\eta_x)^2 + u_\xi\xi_{xx} + u_\eta\eta_{xx} \\ u_{yy} = u_{\xi\xi}(\xi_y)^2 + 2u_{\xi\eta}\xi_y\eta_y + u_{\eta\eta}(\eta_y)^2 + u_\xi\xi_{yy} + u_\eta\eta_{yy} \end{cases} \Rightarrow$$

$$u_{xx} = u_{\xi\xi}(1)^2 + 2u_{\xi\eta}(1)\cdot(0) + u_{\eta\eta}\cdot(0)^2 + u_\xi\cdot(0) + u_\eta\cdot 0 \Rightarrow$$

(5) $$u_{xx} = u_{\xi\xi}$$

$$u_{yy} = u_{\xi\xi}\cdot(0)^2 + 2u_{\xi\eta}(0)\cdot(1) + u_{\eta\eta}(1)^2 + u_\xi\cdot 0 + u_\eta\cdot(0) \Rightarrow$$

(6) $$u_{yy} = u_{\eta\eta}$$

Putting this then into the Laplace eq-n:

$$\Delta u = u_{xx} + u_{yy} = u_{\xi\xi} + u_{\eta\eta} = 0 \Rightarrow$$

(7) $$\boxed{u_{\xi\xi} + u_{\eta\eta} = 0}$$

the defined transf-n does not change the Laplace eq-n. $\Rightarrow$ $u(\xi,\eta)$ satisfies the Laplace eq-n.

by the result of part (i) $(\xi,\eta) \leftarrow (x,y)$ transf-n is OK $\Rightarrow$ $u(x,y) \in C^2(\Omega) \Rightarrow$

$u(\xi,\eta) \in C^2(\Omega')$, $\Omega' = \{(\xi,\eta) : \xi = x+a,\ \eta = y+b,\ \forall (x,y) \in \Omega\}$

as desired.